

Issac Tabares

Knowledge & Systems Experience | Scientific Study | Technical Practice

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This document is a **knowledge and systems record**. It is meant to describe the systems I work with, the scientific and technical topics I study, and the areas of computational, analytical, and creative knowledge that shape the way I think.

I currently work at **BlackLine** as a **Data and Systems Integration Tier 2 Analyst**.

I am an engineer with over **14 years of experience** in enterprise application support, systems integration, identity and access management, data architecture, networking, and cybersecurity across the **financial technology**, **healthcare**, and **telecommunications** sectors. My **Industrial Engineering** background provides a structured approach to root-cause analysis, compliance integrity, cross-functional coordination, and sustained operational stability.

I am drawn to systems because they expose hidden structure. That is true whether I am troubleshooting an enterprise workflow, studying a quantum equation, analyzing data flow, learning a scientific model, composing at the piano, studying anatomy, or modeling a physical phenomenon in 3D. I tend to ask the same questions in different areas: what connects to what, where does the failure originate, what remains stable when perspective changes, and what deeper structure sits underneath the visible behavior.

My technical side and my scientific side are not separate. The same attention I use for **data reliability**, **workflow behavior**, **file validation**, and **systems troubleshooting** is the attention I bring to **quantum mechanics**, **relativity**, **black holes**, **cosmology**, **human biology**, **music**, **computation**, and **visual modeling**.

A lot of my curiosity began with questions about **time**, **existence**, and the origin of everything. Physics gives structure to those questions. **Quantum mechanics** changed how I think about reality beneath appearance. **Relativity** changed how I think about time, measurement, and perspective. **Black holes** make me confront limits: space, causality, information, measurement, and existence. **Cosmology** lets me think about the universe as a history, not only as a place.

Systems and Integration Experience

My current technical work is centered on reliability, structure, and recovery. I work best where systems are interconnected, data must remain trustworthy, and small errors can create large downstream consequences.

In my current role, I serve as the **primary technical escalation point** between frontline support and internal engineering teams, guiding enterprise clients through advanced authentication, provisioning, and integration issues.

- Experience working across **ERP-connected systems, financial platforms**, and enterprise workflows where data consistency must be maintained across multiple integration points
- Analysis of **structured data flows**, including validation of files, formats, system expectations, and downstream processing requirements
- Troubleshooting across **data pipelines, source systems, file structures, transport layers**, and **application-level processing**
- **SFTP-based integration** work involving transmission failures, delivery validation, routing issues, permission handling, and restoration of data flow into downstream systems
- **CSV/TXT validation** using **Notepad++**, including delimiter normalization, encoding verification (**UTF-8 / ANSI**), and format alignment for system-side processing
- **API-level testing and validation** with **Postman**, used to examine connectivity, response behavior, payload structure, and integration-side system issues
- Advanced use of **Excel** for data validation, pattern detection, structural comparison, and practical analytical modeling
- End-to-end management of integration workflows through **NetSuite and SAP connectors** and **direct REST APIs**, including **JSON/XML** structures and **API token authentication**
- Administration of **SFTP/FTP** transfers and **SSH key authentication**, including key handling with **PuTTYgen**

Identity and Access Management (IAM)

After joining BlackLine, I **volunteered to help with an SSO project** and developed practical experience in **SSO and SAML identity federation within my first few months at the company**. I entered the project without prior experience in this area. I gained that experience primarily through **live, real-time meetings with clients**, working face to face over **Microsoft Teams, Cisco Webex, Zoom, Google Meet**, and other conferencing platforms.

During these sessions, I guide both the clients' **SSO administration teams** and their **BlackLine system administrators** through the configuration process **step by step**, coordinating the work across the identity-provider and application sides. This direct work with clients, together with technical investigation and troubleshooting, has developed my experience with authentication flows, federation metadata, identity mappings, certificates, and access controls.

- Live configuration guidance for **Single Sign-On (SSO)** and **SAML 2.0 identity federation** migrations, working directly with client administrators as they execute the required configuration steps
- Coordination between **client SSO administrators** and **client application administrators** to review settings, investigate authentication issues, and validate configuration changes during real-time sessions
- Identity troubleshooting in **multi-tenant environments**, involving **Microsoft Entra ID (Azure AD), Okta, PingFederate**, and **SailPoint**, according to their roles in identity federation, management, and provisioning
- Resolution of advanced **authentication** and **user provisioning** issues, including account lifecycle management and **Multi-Factor Authentication (MFA)**
- Experience with **Active Directory Federation Services (ADFS)** and **Group Policy** in enterprise identity and access environments
- Review of **SAML metadata** and federation relationships, including **Identity Provider (IdP)** and **Service Provider (SP)** roles, entity identifiers, audience values, **Assertion Consumer Service (ACS)** endpoints, and **NameID** mappings
- Analysis of **IdP-initiated and SP-initiated authentication flows**, domain-based routing, and tenant isolation considerations
- Review of **SAML signing certificates**, certificate validity and rotation planning, digital signatures, transport encryption, and **HTTP-POST / HTTP-Redirect bindings**
- Comparison of federation settings across **test and production environments** to identify configuration differences and support migration readiness

SSO Security Controls and Access Governance

My work on the project also developed my practical knowledge of **authentication enforcement**, **access governance**, **session behavior**, and **audit traceability**. I analyze how identity-provider policies and application controls interact, including their relevance to **SOX-related IT General Controls (ITGC)**.

- Analysis of the distinction between **authentication (AuthN)** and **authorization (AuthZ)**, including pre-provisioned accounts, role assignments, and **Just-In-Time (JIT) provisioning** considerations
- Review of **user deactivation and deprovisioning**, distinguishing prevention of new authentication from existing session behavior and application-level account status
- Assessment of **external and guest access** arrangements, including federated identities, least-privilege access, account ownership, and access lifecycle controls
- Analysis of **MFA and conditional access responsibilities** across identity and application layers
- Review of **session lifecycle controls**, including identity-provider re-authentication policies and application idle timeouts
- Evaluation of **service account governance** and the separation between interactive browser authentication and non-interactive integration authentication
- Analysis of **authentication enforcement**, controlled recovery options, authorized configuration changes, and the evidence available to assess these controls
- Review of **audit logging and change traceability**, including administrator accountability and evidence requirements for identity configuration changes
- Consideration of **data minimization** and the handling of identity attributes within federation exchanges

Data, Reliability, and Structured Troubleshooting

The part of technical work that interests me most is the structure underneath a failure. I care about how data moves, how systems interpret it, how formats break, how permissions block flow, how timing changes outcomes, and how a visible issue can originate somewhere deeper in the chain.

- Data validation across file-based and platform-based workflows
- Pattern recognition in repeated incidents, malformed inputs, missing fields, failed transmissions, and inconsistent system behavior
- Root-cause reasoning across multiple layers: user action, file structure, permissions, transmission, platform processing, and downstream interpretation
- Documentation of observed behavior, reproduction steps, suspected cause, actual cause, and resolution path
- Practical understanding of how business operations depend on accurate technical processing, especially when financial data and workflow timing are involved
- Experience working inside high-volume enterprise environments where **accuracy**, **availability**, and **processing reliability** matter directly
- Experience in **secure Remote Desktop (RDP) environments**, including template handling, system validation, and controlled workflow execution
- Additional operational support using **automation and observability tools**, including **New Relic**
- Design of **large-scale data exports** and **compliance extractions**, including handling and sequencing of **multi-part archives with 7-Zip**
- **Jenkins job orchestration** for operational workflows and data movement
- Use of **backend developer tooling**, **server file systems**, and observability platforms to trace data movement and overcome infrastructure bottlenecks
- **SQL database administration** within enterprise environments

Enterprise Platforms and IT Tools

My IT experience includes enterprise support, infrastructure exposure, device ecosystems, access tools, endpoint technologies, service platforms, workflow tools, and structured troubleshooting. The value of this experience is not only tool familiarity; it is the ability to interpret how systems fail, how users experience failure, and how technical causes move across layers.

- Enterprise platforms: **ServiceNow**, **RT (Request Tracker)**, **Jira**, **Salesforce**
- Infrastructure exposure: **IBM Mainframes**, **Citrix**, **VMware**, **Active Directory**, **SCCM**, and **Linux**
- Technical scope including **SQL databases**, **mail server infrastructures**, and enterprise web-based environments using **Visual Studio**
- Familiarity with **SAP Fiori**, **Rivermine (Tangoe)**, and **Coupa**, supporting work across enterprise financial and operational ecosystems
- Working knowledge of **network configuration**, **Cisco**, and **Avaya** technologies
- Cloud and platform exposure including **AWS**, **GCP**, **Azure**, and **Tableau Server**
- Experience with endpoint, security, and access technologies including **Symantec Endpoint Protection**, **McAfee**, **Bitdefender**, **Cisco AnyConnect**, **PingID**, **Dell Data Security**, and **AirWatch**
- Remote support delivery using **BeyondTrust**, **LogMeIn**, **TeamViewer**, and **Splashtop**
- Support across modern device ecosystems including **iOS devices**, **iPads**, **Mi-Fi units**, and enterprise peripherals
- Experience in **Windows** and **macOS** environments, in addition to **Linux**
- Work with **TCP/IP**, **DNS**, **DHCP**, **VPNs**, **VLANs**, **LAN/WAN**, and **firewall management**
- Use of **Salesforce Service Cloud**, **Confluence**, **Microsoft 365**, and **PowerApps** in enterprise service and collaboration ecosystems
- Diagnostics using **Windows Event Logs**

Scientific Topics

My scientific interests are personal before they are academic. I study them because they speak directly to the questions that have followed me for most of my life. They also shape the way I think about systems, models, structure, information, and limits.

- **Classical mechanics**, including forces, motion, energy, action, Lagrangian and Hamiltonian views, orbits, and physical modeling
- **Electromagnetism**, including fields, Maxwell's equations, light, waves, radiation, optics, circuits, and electromagnetic energy flow
- **Thermodynamics and statistical mechanics**, including entropy, temperature, equilibrium, irreversibility, microstates, the arrow of time, and information limits
- **Mathematical foundations**, including calculus, linear algebra, differential equations, probability, geometry, matrices, eigenvectors, and mathematical structure behind physics
- **Quantum mechanics**, especially the wave function, probability, measurement, the Schrodinger equation, atomic structure, superposition, interference, and the idea that reality may be deeper than what is directly visible
- **Quantum entanglement**, because it shows that a whole system can contain structure that is not reducible to isolated parts
- **Quantum computing**, including qubits, gates, superposition, entanglement, algorithms, Grover search, Shor's algorithm, circuit models, and the practical bridge between physics and computation
- **Quantum complexity**, because it asks what quantum systems can compute efficiently, what remains difficult, and how physical structure changes computation

- **Quantum field theory**, including fields, particles as excitations, vacuum structure, propagators, micro-causality, gauge fields, the Higgs field, and the Standard Model
- **Relativity and spacetime**, especially time dilation, Lorentz transformations, spacetime geometry, proper time, gravitational redshift, and the tension between general relativity and quantum theory
- **Black holes**, especially event horizons, singularities, photon spheres, black-hole shadows, black-hole thermodynamics, Hawking radiation, the Page curve, and the unresolved questions around information and existence
- **Wormholes**, including Einstein-Rosen bridges, traversability questions, exotic stress-energy, energy conditions, stability, causality, and the boundary between serious geometry and speculation
- **Cosmology and the early universe**, including the Big Bang, inflation, cosmic expansion, dark ages, first stars, galaxy formation, dark energy, cosmic horizons, possible futures of the universe, and the history of the universe
- **Information theory**, including Shannon entropy, mutual information, channel capacity, coding, compression, KL divergence, Landauer's principle, and the link between information and physics
- **Artificial intelligence**, including neural networks, transformers, attention, gradient descent, optimization, diffusion models, RLHF, calibration, model limits, hallucination risks, and the distinction between AI behavior and conscious interiority
- **Programming and algorithms**, including structured programming, debugging, Dijkstra-style pathfinding, sorting, recursion, dynamic programming, Big O growth, numerical state, and computational logic
- **Numerical methods and simulations**, including Euler methods, Runge-Kutta methods, Monte Carlo methods, heat diffusion, stability, convergence, error, and the danger of attractive visuals hiding numerical failure
- **Chemistry and molecular structure**, including bonding, molecular geometry, reaction rates, equilibrium, Gibbs free energy, spectroscopy, quantum chemistry, materials, batteries, catalysts, and biomolecular structure prediction
- **Biology and life systems**, including DNA, protein structure, information flow, cells, metabolism, evolution, population growth, systems biology, structural biology, and biological organization
- **Human anatomy and biology**, including study through **Complete Anatomy**, body structure, organs, muscles, skeletal systems, motion, and the organization of living form
- **Neuroscience and consciousness**, including neural correlates, cognitive access, subjective experience, recurrent dynamics, leaky integrate-and-fire models, measurement limits, EEG, fMRI, stimulation, lesion studies, and the hard problem of consciousness
- **Philosophy of science**, including explanation, evidence, model limits, scientific realism, interpretation, uncertainty, falsifiability, and the boundary between established science and personal metaphysics
- **Complex systems and emergence**, including networks, feedback, self-organization, phase transitions, ecosystems, brains, economies, social systems, and patterns that appear from many interacting parts
- Continued interest in **quantum chromodynamics, asteroseismology, fluid mechanics, geometric optics, black-hole thermodynamics, astrophysics**, and computational approaches to physical systems

Interpretive Scientific Questions

My interpretive work centers on time, infinity, quantum theory, consciousness, and the question of how finite perspectives exist inside a larger total reality. I separate established physics from interpretation and personal metaphysical commitment.

- **Time**, including proper time, coordinate time, thermodynamic time, quantum time, relational time, semiclassical time, and experienced time
- **Infinity**, including mathematical infinity, physical infinity, ultraviolet divergences, singularities, cosmic infinity, renormalization, regularization, and the distinction between artifacts and incompleteness signals
- **Everettian quantum interpretation**, including unitary evolution, decoherence, branch structure, records, probability, and the reality of branch-relative histories
- **Finite configurations**, including bodies, observers, memories, systems, branches, worlds, records, and local structures inside a wider totality
- **Emergent locality**, including the idea that spatial nearness may arise from information structure, entanglement, mutual information, and Hilbert-space relations
- **Consciousness-first ontology**, understood as a personal metaphysical commitment about first-person interiority, not as a claim that quantum mechanics proves consciousness
- **Generative variation**, including the way reality appears through many species, bodies, minds, stars, worlds, branches, histories, and possible configurations

Computational Development

Computation gives me a way to test, model, visualize, and organize the questions I care about. It connects my technical work with scientific modeling, interface design, and visual explanation.

- Expansion into **Tableau Server**, with focus on enterprise data visualization architecture, performance optimization, and governance
- Continued development in **HTML**, **JavaScript**, and **CSS**, with attention to structure, performance, and interface clarity
- Practical site work using **VS Code**, content structure, interface logic, debugging, accessibility review, and clean file organization
- Continued study of **Python** and **C++**, centered on computational problem-solving, data-oriented workflows, and scientific modeling
- Interest in **Qiskit** and quantum computing tools as a practical bridge between theory, simulation, and future quantum workflows
- Ongoing work with **3D modeling**, **Monte Carlo methods**, Web-based visualization, and visual representations of physical phenomena
- Interest in **interactive educational models**, especially when equations, controls, and visual behavior can make an abstract idea easier to understand
- Experience organizing a large personal website through structured content, bilingual sections, local assets, scientific models, source references, interface states, accessibility controls, and clean file boundaries

Creative Practice

Music and visual work are not separate from the way I think. They are part of the same structure. **Piano composition** gives me a way to turn thought into sound, and composition forces the same kind of discipline I value in systems: tension, restraint, movement, balance, and resolution.

My creative practice also includes **guitar**, **visual design**, **interface structure**, **3D modeling**, and art-driven ways of explaining complex ideas. I care about whether something is technically correct, but I also care about whether it feels coherent, intentional, and alive.

Certifications

- ✓ **ITIL Foundation Certification** – June 2024
- ✓ **Software Security Assurance – HiTech (TCS)** – August 2023
- ✓ **Generative AI Fundamentals (TCS Curriculum)** – February 2024

Education

Instituto Politecnico Nacional